ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024 - 2025)

JEE (Advanced)
TEST # 02
21-07-2024

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Unit Test # 02

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4
	A.	B	B	B	A
SECTION-I (ii)	Q.	5	6	7	8
	A.	B,C,D	A,C	A,C,D	B,C,D
SECTION-I (iii)	Q.	9	10	11	12
	A.	C	C	C	C
SECTION-I (iv)	Q.	13	14	15	16
	A.	A	B	A	C
SECTION-II	Q.	1	2	3	4
	A.	6.25	34.00	1.00	0.50

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4
	A.	A	A	C	C
SECTION-I (ii)	Q.	5	6	7	8
	A.	A,B,C	A,C	A,B,C,D	A
SECTION-I (iii)	Q.	9	10	11	12
	A.	A	B	B	B
SECTION-I (iv)	Q.	13	14	15	16
	A.	A	D	C	C
SECTION-II	Q.	1	2	3	4
	A.	9.00	8.00	7.00	4.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4
	A.	A	C	D	B
SECTION-I (ii)	Q.	5	6	7	8
	A.	B,C	A,B,C	A,C	C,D
SECTION-I (iii)	Q.	9	10	11	12
	A.	C	B	B	D
SECTION-I (iv)	Q.	13	14	15	16
	A.	C	B	A	C
SECTION-II	Q.	1	2	3	4
	A.	84.00	1.00	4.00	10.00

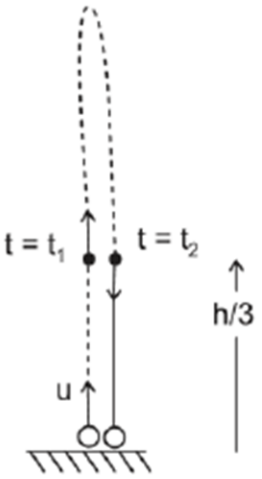
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HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (B)**



Max. Height = $h = \frac{u^2}{2g}$

$\Rightarrow u = \sqrt{2gh}$

$S = ut + \frac{1}{2}at^2$

$\frac{h}{3} = \sqrt{2gh}t + \frac{1}{2}(-g)t^2$

$\frac{gt^2}{2} - \sqrt{2gh}t + \frac{h}{3} = 0$ (Roots are t_1 & t_2)

$$\frac{t_2}{t_1} = \frac{\sqrt{2gh} + \sqrt{2gh - 4 \times \frac{g}{2} \times \frac{h}{3}}}{\sqrt{2gh} - \sqrt{2gh - 4 \times \frac{g}{2} \times \frac{h}{3}}}$$

$$= \frac{\sqrt{2gh} + \sqrt{\frac{4gh}{3}}}{\sqrt{2gh} - \sqrt{\frac{4gh}{3}}} = \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$$

2. **Ans (B)**

$S_{th} = u + \frac{1}{2}a(2n - 1)$

$S_{th} = 9 + \frac{1}{2}(-2)(2 \times 5 - 1)$

$S_{th} = 9 - 9$

displacement in fifth second = 0

$v = u + at$

$0 = 9 - 2t$

$2t = 9$

$t = 4.5$ sec

t (0, 5)

distance covered in fifth second

$$S = ut + \frac{1}{2}at^2 + \frac{1}{2} \times 2 \times \left(\frac{1}{2}\right)^2$$

$$= 0 + \frac{1}{2} \times 2 \times \left(\frac{1}{2}\right)^2 + \frac{1}{2} \times 2 \times \frac{1}{4}$$

$$= \frac{1}{2} \times 2 \times \frac{1}{4} = 0.25 + 0.25 = 0.50\text{m}$$

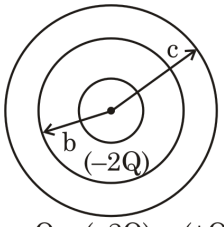
3. **Ans (B)**

Self leaving flux on each face = ϕ'

$$\phi + \phi' = \frac{Q}{\epsilon_0} \Rightarrow \phi = \phi' = \frac{Q}{2\epsilon_0}$$

$$F = \sigma \int EdA \cos\theta = \sigma \frac{Q}{2\epsilon_0} = \frac{Q^2}{2\epsilon_0 d^2}$$

4. **Ans (A)**



$-Q - (-2Q) = (+Q)$

Charge induced on inner surface of shell = $-2Q$

Charge induced on outer surface of shell

$= -Q - (-2Q) = Q$

$$\text{Surface charge density } (\sigma) = \frac{\text{Charge}}{\text{Area of surface}}$$

$$\sigma_{(b)} = \frac{-2Q}{4\pi b^2} \; ; \; \sigma_{(c)} = \frac{Q}{4\pi c^2}$$

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PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,C,D)

Component of velocity in vertical direction is same (H_{max} is same)

$v_1 \sin\theta_1 = v_2 \sin\theta_2$ ....(1)

(A) Time period

$T_A = \frac{2u_1 \sin\theta_1}{g}$

$T_B = \frac{2u_2 \sin\theta_2}{g}$

$T_A = T_B$

(B) $R_A \Rightarrow \frac{2u_1^2 \sin\theta, \cos\theta_1}{g}$

$R_B \Rightarrow \frac{2u_2^2 \sin\theta_2, \cos\theta_2}{g}$

$R_B > R_A$

from eq^n – 1

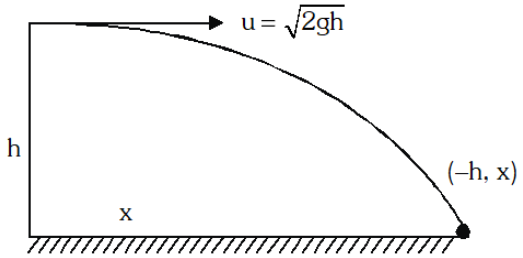
$u_1 \cos\theta_1 < u_2 \cos\theta_2$

since H_{max} is same

$u_1 \sin\theta_1 = u_2 \sin\theta_2$

6. Ans (A,C)

Using equation of trajectory :



$-h = x\tan(0°) - \frac{gx^2}{2(2\,gh)(\cos^2 0°)} \Rightarrow x = 2h$

PART-1 : PHYSICS

SECTION-I (iii)

10. Ans (C)

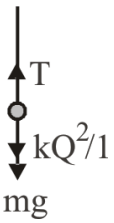
$\frac{H}{2} = \frac{v_y^2}{2g}$ where v_y is velocity at $\frac{H}{2}$ maximum height & v_y = 20 m/s, so H = 40 m.

11. Ans (C)

$mg4 + \frac{kQ^2}{5} - \frac{kQ^2}{1} = 0$

$m = \frac{4 \times 9 \times 10^9 \times (20)^2 \times 10^{-12}}{5 \times 40} = 72$ gm

12. Ans (C)



v = 0 at bottom

$T = mg + \frac{kQ^2}{1}$

T = 0.72 + 3.6 = 4.32 N

PART-1 : PHYSICS

SECTION-I (iv)

13. Ans (A)

For (P) :

At t = 2s, $x = 2(2)^3 - 15(2)^2 + 36(2) = 28$m

For (Q) :

As velocity changes its sign at t = 2s and

t = 3s so for $0 \le t \le 2$ sec distance = |displacement|

Therefore Distance travelled = 28m

For (R) :

$v = \frac{dx}{dt} = 6t^2 - 30t + 36 = 6(t-2)(t-3)$

At t = 5s, $v = 6(5-2)(5-3) = 36$ ms^{-1}

For (S) :

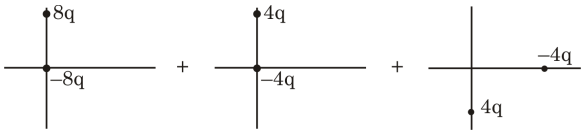
Acceleration $a = \frac{dx}{dt} = 12t - 30$,

At t = 5s, a = 30 ms^{-2}

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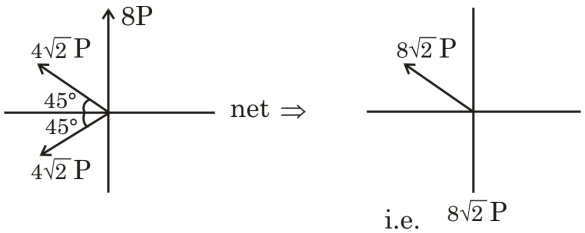
15. **Ans (A)**

(P) Consider it as



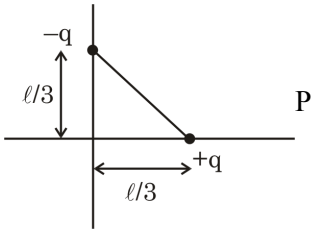
Let P = Ql

than we have



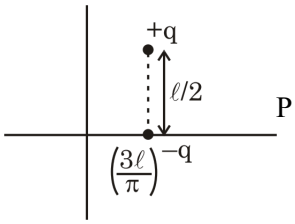
i.e. $8\sqrt{2}\,P$

(Q) We can assume charge of cone to be at point $\left(0, \frac{1}{3}\right)$



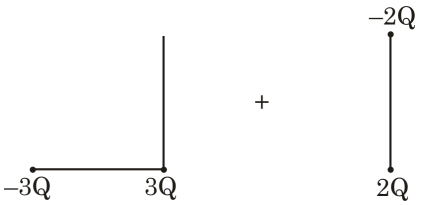
$$P = \frac{Q\ell\sqrt{2}}{3}$$

(R) We can assume charge -q at point $\left(\frac{3l}{\pi}, 0\right)$

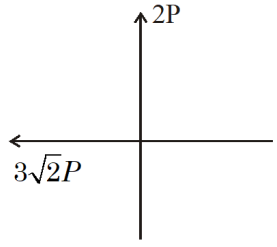


$$P \Rightarrow \frac{q\ell}{2}$$

(S) Assume as



We have P = Ql



Let = $\sqrt{22}$ P

$\sqrt{22}$ Ql

16. **Ans (C)**

total flux through cube

$$= \frac{Q}{4\varepsilon_0} + \frac{2Q}{\varepsilon_0} + \frac{3Q}{8\varepsilon_0} = \frac{21Q}{8\varepsilon_0}$$

flux through surface ABCD

$$= 0 + 0 + \frac{2Q}{6\varepsilon_0} = \frac{Q}{3\varepsilon_0}$$

flux through surface ABFE

$$= 0 + \frac{2Q}{6\varepsilon_0} + \frac{3Q}{24\varepsilon_0} = \frac{11Q}{24\varepsilon_0}$$

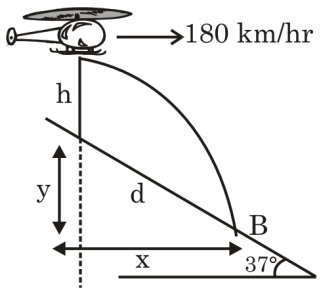
Sum of flux through BCGF and EFGH

$$= \frac{Q}{8\varepsilon_0} + 2 \times \frac{2Q}{6\varepsilon_0} + 2 \times \frac{3Q}{24\varepsilon_0} = \frac{25Q}{24\varepsilon_0}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (6.25)**



v = 180 km/hr = 50 m/s

x = 50 (10) = 500 m

$$\tan 37 = \frac{y}{x} \Rightarrow \frac{3}{4} = \frac{y}{500}$$

y = 375 m

$$d = \sqrt{x^2 + y^2} = 625\text{m}$$

$$(y + h) = \frac{1}{2}gt^2 = \frac{1}{2} \times 10(10)^2 = 500\text{m}$$

h = 125 m

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2. **Ans (34.00)**

$$\frac{H}{R} = \frac{\tan\theta}{4} \Rightarrow \tan\theta = 4$$

$$\Rightarrow \sin\theta = \frac{4}{\sqrt{17}}$$

$$H_{max} = \frac{u^2\sin^2\theta}{2g} = \frac{8u^2}{17g}$$

3. **Ans (1.00)**

$$\frac{nQ}{5\varepsilon_0} = \frac{3Q}{5\varepsilon_0} - \left[\frac{Q}{\varepsilon_0} - \frac{3Q}{5\varepsilon_0}\right] = \frac{Q}{5\varepsilon_0}$$

$n = 1$

4. **Ans (0.50)**

E_x due to $\lambda_1 = E_x$ due to λ_2

$$\frac{2k\lambda_1}{R} = \frac{k\lambda_2}{R} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{1}{2}$$

PART-2 : CHEMISTRY

SECTION-I (i)

4. **Ans (C)**

$$\frac{\ln 2}{t_{1/2}}(10) = \ln\left(\frac{80}{20}\right)$$

$t_{1/2} = 5$ minute

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,B,C)**

nodal plane of p_x is yz the lobes of d_{xy} present in xy plane.

7. **Ans (A,B,C,D)**

For experiment no. 2 and 3, [B] = constant

$$\frac{(A_0)_2}{(A_0)_3} = \frac{0.04}{0.01} = \frac{4}{1} \text{ and also,}$$

$$\frac{r_2}{r_3} = \frac{1.6\times10^{-2}}{4\times10^{-3}} = \frac{4}{1}$$

Rate $\propto [A]^1$ i.e. the reaction is of first order w.r.t. A.

For experiment no. 1 and 2.

$$\frac{r_1}{r_2} = \frac{K[A_0]_1^1\,[B_0]_1^x}{K[A_0]_3^1\,[B_0]_2^x}$$

$$\frac{4\times10^{-3}}{1.6\times10^{-2}} = \frac{0.02\,(0.03)^x}{0.04\,(0.06)^x}$$

Solving, x = 1 or rate $\alpha[B]^1$

i.e. reaction is of first order w.r.t. B also

$\therefore$ Rate $= K[A_0]^1\,[B_0]^1$

(A) Correct

The mechanism of reaction not provided, so it may be elementary.

(B) Correct

Rate = K[A] [B]

using data of experiment no.1, 4×10^{-3}

$= K[0.02]\,[0.03]$

$\Rightarrow K = 6.67\ m^{-1}\ s^{-1}$.

(C) Correct

The mechanism of reaction is not provided so it may be complex.

(D) Correct

Rate, $r_1 = K[A]\,[B] = 4\times10^{-3}$

Volume is doubled $\Rightarrow$ concentration is halved.

$$\therefore r_1^1 = K\left[\frac{A}{2}\right]\left[\frac{B}{2}\right] = \frac{K[A]\,[B]}{4} = \frac{r}{4} = \frac{4\times10^{-3}}{4} = 10^{-3}$$

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8. **Ans (A)**

$$E_{a(overall)} = \frac{2}{3}(E_{a_1} + E_{a2} - E_{a3})$$

$$= \frac{2}{3}(180 + 80 - 50)$$

$$= 140 \text{ kJ/mol}$$

PART-2 : CHEMISTRY

SECTION-I (iii)

9. **Ans (A)**

$n = 2 \longrightarrow \ell = 0 \quad 1 \quad 2 \quad 3$

$\quad\quad\quad

3. **Ans (7.00)**

For $T_{75\%} = A\,T_{50\%}$ Let initial conc. = 100

$T_{75} = \frac{2.303}{K}\log\frac{100}{25}$ $a-x = 100 - 75 = 25$

$T_{75} = \frac{2.303}{K}2\log 2$

For T_{50}

$T_{50} = \frac{2.303}{K}\log\frac{100}{50}$ $a-x = 100 - 50 = 50$

$T_{50} = \frac{2.303}{K}\log 2$

$\frac{T_{75}}{T_{50}} = \frac{\frac{2.303}{K}\times 2\times\log 2}{\frac{2.303}{K}\times\log 2}$

$\boxed{T_{75\%} = 2T_{50\%}}$ So A = 2

Now $T_{87.5\%} = B \times T_{50\%}$

$T_{87.5} = \frac{2.303}{K}\log\frac{100}{12.5}$

$T_{87.5} = \frac{2.303}{K}\log 8$

$T_{87.5} = \frac{2.303}{K}\times 3\log 2$

Divide $\boxed{T_{87.5} = 3T_{50}}$ So $\boxed{B = 3}$

Now, $T_{99\%} = CT_{90\%}$

$T_{99} = \frac{2.303}{K}\log\frac{100}{1}$

$T_{99} = \frac{2.303}{K}\times 2$

$T_{90} = \frac{2.303}{K}\log\frac{100}{10}$

$T_{90} = \frac{2.303}{K}$

$\frac{T_{99}}{T_{90}} = \frac{\frac{2.303}{K}\times 2}{\frac{2.303}{K}}$

$\Rightarrow T_{99} = 2T_{90}$

$C = 2$

$\Rightarrow A + B + C$

$\Rightarrow 2 + 3 + 2 = 7$

4. **Ans (4.00)**

$t = \frac{1}{0.693}\times\ln\frac{100}{6.25} \Rightarrow t = \frac{0.1}{0.693}\ln 16$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**

$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$f'(x) = \lim_{h\to 0}\frac{f(x)\,f(1+h/x)-f(x)}{h}$

$f'(x) = \frac{f(x)}{x}f'(1)$

$\frac{f'(x)}{f(x)} = \frac{4}{x} \Rightarrow f(x) = x^4$

2. **Ans (C)**

$\lim (\cos x + a^3\sin(b^6x))^{1/x}$

$= e^{\lim_{x\to 0}\frac{\cos x + a^3\sin b^6x - 1}{x}}$

$e^{\lim_{x\to 0}\left(\frac{\cos x-1}{x}+\frac{a^3\sin b^6x}{x}\right)}$

$e^{0+a^3b^6} = e^{a^3b^6}$

$a^3b^6 = 512$

$ab^2 = 8$

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3. **Ans (D)**

$$\lim_{x\to0^-} f(x) = \lim_{x\to0}\left(\frac{\sin(a+2)x}{x} + \frac{\sin x}{x}\right) = a+3$$

$$\lim_{x\to0^-} f(x) = \lim_{x\to0} \frac{(x+3x^2)^{1/3} - x^{1/3}}{x^{4/3}}$$

$$= \lim_{x\to0} \frac{(1+3x)^{1/3} - 1}{x} = 1$$

$f(0) = b$

for continuity at x = 0

$$\lim_{x\to0^-} f(x) = f(0) = \lim_{x\to0^+} f(x)$$

$\Rightarrow \quad a + 3 = b = 1$

$\therefore \quad a = -2, \quad b = 1$

$\therefore \quad a + 2b = 0$

4. **Ans (B)**

$$\log_{\cos x}\left(\frac{2\sin x + 3}{4}\right) \geqslant 2, \ x \in [0, \pi]$$

$$\Rightarrow \quad \frac{2\sin x + 3}{4} \leqslant \cos^2 x$$

$$\Rightarrow \quad 4\sin^2 x + 2\sin x - 1 \leq 0$$

$$\Rightarrow \quad \left(\sin x - \left(\frac{1+\sqrt{5}}{4}\right)\right)\left(\sin x - \left(\frac{-1-\sqrt{5}}{4}\right)\right) \leqslant 0$$

$$\frac{-1-\sqrt{5}}{4} \leqslant \sin x \leqslant \frac{1+\sqrt{5}}{4}$$

$$0 \leqslant \sin x \leqslant \frac{1+\sqrt{5}}{4}$$

$$x \in \left(0, \frac{\pi}{10}\right]$$

Ans. (B)

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B,C)**

(A) $f(x) = \{x\}\sin^2 \pi x$ differentiable $\forall\ x \in R$

(B) Not differentiable at $x = (2n + 1)\pi$

(C) Not differentiable at x = 0, 2, 6, 4

(D) Not differentiable at all integers

6. **Ans (A,B,C)**

$f(2) = 0$

$$f(2^+) = \lim_{h\to0} \cos\left[\frac{\pi}{2+h}\right] \cos\left(\frac{\pi}{2}(1+h)\right)$$

$$= -\lim_{h\to0} \cos\left[\frac{\pi}{2+h}\right] \sin\frac{\pi}{2}h = 0$$

$$f(2^-) = \lim_{h\to0} \cos\left[\frac{\pi}{2-h}\right] \sin\frac{\pi}{2}h = 0$$

$f(1) = \cos[p] \cos 0 = \cos 3$

$$f(1^+) = \lim_{h\to0} \cos\left[\frac{\pi}{1+h}\right] \cos\frac{\pi}{2}h = \cos 3 = f(1^-)$$

$f(0) = 0$

$f(0^+) = 0$

$f(0^-) = 0$

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7. **Ans (A,C)**

$5^{\log_a x} + 5x^{\log_x 5 . \log_a x} = 3$

$\Rightarrow 5^{\log_a x} + 5.5^{\log_a x} = 3$

$\Rightarrow 6.5^{\log_a x} = 3$

$\Rightarrow 5^{\log_a x} = \frac{1}{2}$

$\Rightarrow \log_a x = \log_5 \left(\frac{1}{2}\right)$

$\log_a x = -\log_5 2$

$\Rightarrow x = a^{-\log_5 2}$

$x = 2^{-\log_5 a}$

$* \log_a 5 = \frac{\log_x 5}{\log_x a}$

$= \log_x 5 \log_a x$

$* a^{\log_c b} = b^{\log_c a}$

8. **Ans (C,D)**

$\lim\limits_{x\to 0} f(x) = \lim\limits_{x\to 0} \frac{x.2^x - x}{1 - \cos x}$

$= \lim\limits_{x\to 0} \frac{x^2}{1-\cos x} . \frac{2^x - 1}{x} = 2.\ln 2 = \ln 4$

$\lim\limits_{x\to\infty} g(x) = \lim\limits_{x\to\infty} 2^x \sin\left(\frac{\ln 2}{2x}\right)$

$= \lim\limits_{x\to\infty} \frac{\sin\left(\frac{\ln 2}{2^x}\right)}{\left(\frac{\ln 2}{2^x}\right)} . \ln 2 = \ln 2$

PART-3 : MATHEMATICS

SECTION-I (iii)

9. **Ans (C)**

$\sin x = \sin \frac{\pi x}{180} \qquad (x > 0)$

$x = k\pi + (-1)^k \frac{\pi x}{180}$

$k = 1$

$x = \pi - \frac{\pi x}{180} \Rightarrow x = \frac{180\pi}{180+\pi}$

$k = 2$

$x = 2\pi + \frac{\pi x}{180} \Rightarrow x = \frac{360\pi}{180-\pi}$

$m = 360,\ n = 180,\ p = 180,\ q = 180$

10. **Ans (B)**

$\lim\limits_{t\to\infty} \sqrt{At^2 + Bt} - Ct = L,\ L = 2$

$\lim\limits_{t\to\infty} \frac{(A - C^2)t^2 + Bt}{t\left(\sqrt{A + \frac{B}{t}} + C\right)} = L$

$A = C^2$ and $\frac{B}{\sqrt{A} + C} = 2$

$\boxed{C = \sqrt{A}}$ $\boxed{B = 4C}$

11. **Ans (B)**

$2 \le \log_5 N < 3 \Rightarrow 25 \le N < 125$

$3 \le \log_3 N < 4 \Rightarrow 27 \le N < 81$

$\therefore 27 \le N < 81$

Maximum integral value of N is 80

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12. **Ans (D)**

$3 \le \log_5 N < 4$

$\Rightarrow 125 \le N < 625$

$4 \le \log_3 N < 5$

$\Rightarrow 81 \le N < 243$

$\therefore 125 \le N < 243$

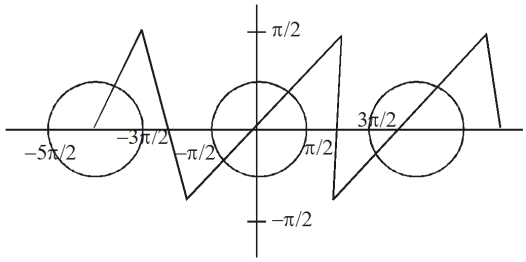
No. of integral values = 118

PART-3 : MATHEMATICS

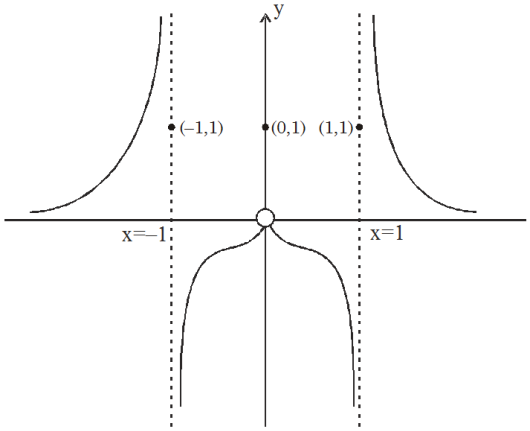
SECTION-I (iv)

13. **Ans (C)**

(P)



(Q)



(R) $\lim\limits_{x\to 0} \left(1 + x\ln\left(1+b^2\right)\right)^{1/x} = 2b\sin^2\theta,\ b > 0$

$\Rightarrow 1 + b^2 = 2b\sin^2\theta \Rightarrow b^2 - 2b\sin\theta + 1 = 0$

$D \geqslant 0 \Rightarrow \sin^4\theta = 1 \Rightarrow \sin\theta = \pm 1$

$\Rightarrow \theta = \pm\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$

$\therefore$ Sum of all possible value of $\theta = 4\pi$

$\therefore k = 4$

(S) for a limit of the function to exist $\alpha + \beta = 1$

since function gain form of 1^{∞} we get $\alpha = -3, \beta = 4$

$\therefore (4\beta + 3\alpha) = 7 \Rightarrow Q$

14. **Ans (B)**

(P) $2n - 1 = 3m - 1$

$2n = 3m$

$\frac{m}{n} = \frac{2}{3}$ $\therefore$ m = 2k, n = 3k and common terms

for k = 1 to 26

(Q) $10\left(\frac{1}{1-a}\right)\left(\frac{1}{1-b}\right) = \frac{10}{1-(a+b)+ab}$

$= \frac{10}{1-\frac{4}{11}-\frac{2}{11}} = \frac{110}{5} = 22$

(R) $S_{20} = S_{10}$ $\therefore S_{30} = 0$

(S) $\frac{3}{1^2.2^2} + \frac{5}{2^2.3^2} \ldots\ldots + \frac{199}{99^2.100^2}$

$= \left(\frac{1}{1} - \frac{1}{2^2}\right) + \left(\frac{1}{2^2} - \frac{1}{3^2}\right) + \ldots\ldots + \left(\frac{1}{99^2} - \frac{1}{100^2}\right)$

$\Rightarrow 1 - \frac{1}{100^2} = \frac{9999}{10000}$

Hence $20|b - a| = 20(1) = 20$

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15. Ans (A)

$m^{(\log_3 5)^2} + n^{(\log_5 7)^2}$

$= \left(m^{\log_3 5}\right)^{\log_3 5} + \left(n^{\log_5 7}\right)^{\log_5 7}$

$= 9^{\log_3 5} + 25^{\log_5 7}$

$= 3^{2\log_3 5} + 5^{2\log_5 7}$

$= 3^{\log_3(25)} + 5^{\log_5(49)} = 25 + 49 = 74$

Again

$m^{\log_3 5} = 9$ & $n^{\log_5 7} = 25$

$\Rightarrow \log_3\left(m^{\log_3 5}\right) = \log_3 9$	$\Rightarrow \log_5\left(n^{\log_5 7}\right) = \log_5(25)$
$\Rightarrow (\log_3 5)(\log_3 m) = 2$	$\Rightarrow (\log_5 7)(\log_5 n) = \log_5(25)$
$\Rightarrow \log_3 m = \dfrac{2}{\log_3 5}$	$= 2$
$= 2\log_5 3$	$\Rightarrow \log_5 n = \dfrac{2}{\log_5 7}$
	$= 2\log_7 5$

$\therefore (\log_3 m)(\log_5 n) = (2\log_5 3)(2\log_7 5)$

$= 4\dfrac{\log 3}{\log 5} \times \dfrac{\log 5}{\log 7}$

$= 4\log_7 3 = \log_7(3^4)$

$= \log_7(81) = \log_b(a)$

$\therefore a = 81$ & $b = 7$

Also, $\text{Antilog}_{\sqrt{b}}(\log_b 2401)$

$= \text{antilog}_{\sqrt{7}}(\log_7 2401)$

$= \text{antilog}_{\sqrt{7}}(4) = \left(\sqrt{7}\right)^4 = 7^2 = 49$

16. Ans (C)

(P) $\dfrac{S_{2n}}{S_n} = \dfrac{\frac{2n}{2}[2a + (2n-1)d]}{\frac{n}{2}[2a + (n-1)d]}$

$= 3 = \dfrac{2a + (2n-1)d}{2a + (n-1)d} = \dfrac{3}{2}$ $\quad \therefore d = \dfrac{2a}{n+1}$

(Q) $S_\infty = 2^{1/4} .4^{1/8} .8^{1/16} \ldots\ldots\ldots\ldots\infty$;

$S_\infty = 2^{1/4} .2^{2/8} .2^{3/16} \ldots\ldots\ldots\ldots\infty$

$S_\infty = 2^{1/4 + 2/8 + 3/16} \ldots\ldots\ldots\infty = 2^{S'_\infty}$

Let $S'_\infty = \dfrac{1}{4} + \dfrac{2}{8} + \dfrac{3}{16} \ldots\ldots\ldots$

$S'_n = \dfrac{1}{4} + \dfrac{2}{8} + \dfrac{3}{16} \ldots\ldots\ldots \dfrac{n}{2^{n+1}}$ ...(i)

$\dfrac{S'_n}{2} = \dfrac{1}{8} + \dfrac{2}{16} + \ldots\ldots\ldots + \dfrac{n-1}{2^{n+1}} + \dfrac{n}{2^{n+2}}$ ...(ii)

(i) − (ii) we get

$\dfrac{S'_n}{2} = \dfrac{1}{4} + \dfrac{1}{8} + \dfrac{1}{16} + \ldots\ldots\ldots \dfrac{1}{2^{n+1}} - \dfrac{n}{2^{n+2}}$

$\Rightarrow \dfrac{S'_n}{2} = \dfrac{1}{4}\left(\dfrac{1 - (1/2)^n}{1 - 1/2}\right) - \dfrac{n}{2^{n+2}}$

$S'_n = \dfrac{2.2}{4}\left(1 - \left(\dfrac{1}{2}\right)^n\right) - \dfrac{2n}{2^{n+2}}$

$\Rightarrow S'_\infty = 1$ $\quad \therefore S_\infty = 2^{S'_\infty} = 2$

(R) x,y,z are in A.P $y = \dfrac{x+z}{2}$, or ; $2y = x + z$

$(x + 2y - z)\,(x + z + z - x)\,(z + x - y)$

$= (x + (x + z) - z)\,(x + z + z - x)\,(2y - y)$

$= 2x .2z .y = 4xyz$

$\therefore k = 4$

(S) $d = \dfrac{31 - 1}{m + 1} = \dfrac{30}{m + 1}$; $\dfrac{A_7}{A_{m-1}}$

$= \dfrac{1 + 7\frac{30}{m+1}}{1 + (m-1)\frac{30}{m+1}} = \dfrac{5}{9}$

$\Rightarrow \dfrac{m + 211}{31m - 29} = \dfrac{5}{9} \Rightarrow 146\,m = 2044$

$\Rightarrow m = 14$ $\quad \therefore \dfrac{m}{7} = 2$

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PART-3 : MATHEMATICS

SECTION-II

1. Ans (84.00)

$$\frac{1}{16}\sum_{r=1}^{n}\frac{\left(16r^4-1\right)+1}{(2r-1)(2r+1)}$$

$$=\frac{1}{16}\sum_{r=1}^{n}\left(4r^2+1\right)+\frac{1}{32}\sum_{r=1}^{n}\left(\frac{1}{2r-1}-\frac{1}{2r+1}\right)$$

$$=\frac{1}{4}\frac{n(n+1)(2n+1)}{6}+\frac{n}{16}+\frac{1}{32}\left(1-\frac{1}{2n+1}\right)$$

$$=\frac{n^3}{12}+\frac{n^2}{8}+\frac{5n}{48}+\frac{n}{16(2n+1)}$$

2. Ans (1.00)

$$=\lim_{x\to -1}\frac{\sqrt{(3x^2+6x+7)}+\sqrt{(5x^2+10x+14)}+2x-3}{\left(\sin\left(\cos^{-1}x\right)\right)^2}$$

$$=\lim_{x\to -1}\frac{\sqrt{(3x^2+6x+7)}+\sqrt{(5x^2+10x+14)}+2x-3}{\left(\sin\sin^{-1}(1-x^2)^{\frac{1}{2}}\right)^2}$$

$\left(\frac{0}{0}\text{form}\right)$

Now differentiate numerator and denominator w.r. to x.

$$=\lim_{x\to -1}\frac{\frac{(6x+6)}{2\sqrt{3x^2+6x+7}}+\frac{(10x+10)}{2\sqrt{5x^2+10x+10}}+2}{-2x}=\frac{2}{2}=1$$

3. Ans (4.00)

$$=e^{\frac{\cos x}{2}+\frac{3}{2^2}\cos x+\frac{5}{2^3}\cos x+\ldots\infty}$$

$$=e^{\cos x\left(\frac{1}{2}+\frac{3}{2^2}+\frac{5}{2^3}+\ldots\infty\right)}$$

Let $s=\frac{1}{2}+\frac{3}{2^2}+\frac{5}{2^3}\ldots\ldots\infty$

$s=3$

$$f(x)=\ln\left(e^{3\cos x}\right)=3\cos x$$
$$g(x)=\left[\frac{1}{3}f(x)\right]=[\cos x]$$

Discontinuous at $x=0,\frac{\pi}{2},\frac{3\pi}{2},2\pi$

4. Ans (10.00)

Let $t_n=1-\frac{2}{n(n+1)}=\frac{n^2+n-2}{n(n+1)}=\frac{(n+2)(n-1)}{(n+1)n}$

$$\prod_{n=2}^{n}t_n=\frac{4}{3}\cdot\frac{5}{4}\cdot\ldots\ldots\frac{(n+2)}{(n+1)}\cdot\frac{1}{2}\cdot\frac{2}{3}\cdot\ldots\ldots\frac{(n-1)}{n}$$

$$\lim_{n\to\infty}\prod_{n=2}^{n}t_n=\lim_{n\to\infty}\frac{1}{3}\cdot\frac{(n+2)}{n}=\frac{1}{3}$$

Thus $P_n=\left(\frac{1}{3}\right)^2=\frac{1}{9}=\frac{a}{b}$

$a+b=10$

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